

# EP363/MP363 Quantum Mechanics 1

## Exercise sheet 1 : Linear algebra

1. (a) Look up and report the definition of
  - i. the hermitean conjugate of a matrix
  - ii. a hermitean or self-adjoint matrix
  - iii. a unitary matrix

(You may try any textbook on quantum mechanics or linear algebra, Wikipedia, or any other source you can find.)

- (b) Determine whether each the following matrices is hermitean:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} i & 1-i \\ 1-i & -i \end{pmatrix}, \quad B = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & 2-i & 0 \\ 2+i & -1 & 1 \\ 0 & 1 & -1 \end{pmatrix}.$$

- (c) Compute the matrix products  $IA, AI, AB$  and  $BA$  with  $I, A$  and  $B$  given above.
- (d) Determine whether each of the following matrices is unitary:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} i & -1-i \\ 1-i & -i \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & i \\ 0 & -i & 0 \end{pmatrix}.$$

2. The matrices  $A, B$  and the vectors  $u, v$  are given by

$$A = \begin{pmatrix} 2 & 0 & -1 \\ 0 & -1 & -2i \\ -1 & 2i & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 0 & 2i \\ 1 & -2i & 3 \end{pmatrix}, \quad u = \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}, \quad v = \begin{pmatrix} i \\ 2 \\ -1 \end{pmatrix}.$$

Compute the commutator  $[A, B]$  and the matrix products  $u^\dagger v, vu^\dagger, u^\dagger Av$  and  $v^\dagger Bv$ , where  $u^\dagger$  denotes the hermitean conjugate of  $u$ .

3. (a) Write the following sets of equations on matrix form. You are not expected to try to solve them.
  - i.

$$2x_1 - 3x_2 - x_3 = 2$$

$$x_1 + 4x_2 + 2x_3 = 3$$

$$x_1 - x_2 + x_3 = 5$$

- ii.

$$x_1 - 2x_2 - x_3 + 3x_4 = 10$$

$$2x_1 + 3x_2 + x_4 = 8$$

$$x_1 - 4x_3 - 2x_4 = 3$$

$$-x_2 + 3x_3 + x_4 = -7$$

(b) Determine whether or not the inverses of the following matrices exist:

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 3 & -1 & 2 \\ 0 & 1 & 0 \\ -3 & 3 & -2 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 3 & -1 \\ 2 & 0 & 4 \\ -2 & 2 & -2 \end{pmatrix}.$$

(c) For each of the matrices below, determine all the eigenvalues and an eigenvector corresponding to each eigenvalue.

$$\begin{pmatrix} 2 & 5 \\ 5 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & 1+2i \\ 1-2i & -2 \end{pmatrix}, \quad \begin{pmatrix} 2 & 1 & 1 \\ 1 & 3 & 2 \\ -1 & 1 & 2 \end{pmatrix}, \quad \begin{pmatrix} 3 & 0 & 3 \\ 0 & 3 & 3 \\ 2 & 3 & 1 \end{pmatrix}.$$